LEONARDO



PRODUCT VERSION

This technical data sheet is related to following product version of Leonardo display:

	LEONARDO DISPLAY STANDARD
PN#	6100549/3x
Standars contents	
Power supply 8-32 Vdc	X
2 CAN bus	X
1 LIN	X
2 USB	X
1 Ethernet BroadR-Reach	X
2 analog video inputs	X
4 digital inputs	X
2 analog inputs	X
2 high side driver outputs	X
1 Vss (Vref)	X
1 light sensor	X
1 speaker	X
1 audio out	X
RTC	X
1 bicolor led	X
Additional contents	
1 microphone input	
1 wi-fi	
1 bluetooth	
Certifications	
Type approval[1] ECE ONU reg10-05 E3	X
Additional certifications	
MTA Speedy Creator ID[2]	8003

GENERAL DESCRIPTION

Leonardo is a display designed for agricultural or off-highway vehicles, on wheels or tracks. This state-of-the-art device features a number of high speed interfaces towards typical automotive communication channels such as CAN, BroadR-Reach and LIN. In addition to a high resolution TFT display with capacitive touch screen and best-in-class brightness and contrast ratio, Leonardo delivers industry standard Android® operating system and a state-of-the-art software tool developed by MTA (MTA Speedy Creator®), which provides an open and easy to use application development environment. The hardware is designed according to the most robust automotive industry standards.

Leonardo main hardware features:

- extended power supply range from 8 to 32 Vdc;
- 12.1" TFT LCD 1280 x 800 pixel;
- capacitive touch screen, with light glove operability;
- 4 digital high side or low side inputs, two with wake capabilities;
- 2 analog inputs (16 or 32 Vdc user definable dynamic range);
- 2x 350 mA high side outputs, with current sense and diagnostics;
- stereo audio output (needs external amplifier);
- 1 voltage output for sensor supply (5 V, 400 mA);
- 1 real time clock;
- 1 mono speaker;
- 1 ambient light sensor;
- 1 dual color status LED;
- 2 analog video inputs on 2 dedicated Mini BNC connectors;
- 1 LIN 2.2 Master bus;
- 1 RS232;
- 2 CAN bus;
- 1 BroadR-Reach Ethernet;
- 2 USB:
  - 1 USB accessible on a type 'A' side connector;
  - 1 USB available on the rear Tyco Superseal connector;

Memory:

- DDR3 RAM 1 GB;
- Flash EMMC 16 GB;
- SERIAL FLASH 16 MB;
- EEPROM 32 kB.

1. The certification is only for the PN#s referenced in this document. Please refer to customer drawings for PN#s derived by these ones (MTA off-the-shelf dashboard), in which custom information are included (certification, label layout...).

2. The "MTA Speedy Creator ID" here listed are referred to the MTA Speedy Creator variants, for display Leonardo product family, written on the product label. Please contact MTA for any info about it.

Software options:

- Android Studio + MTA Speedy Creator and MTA Gate.

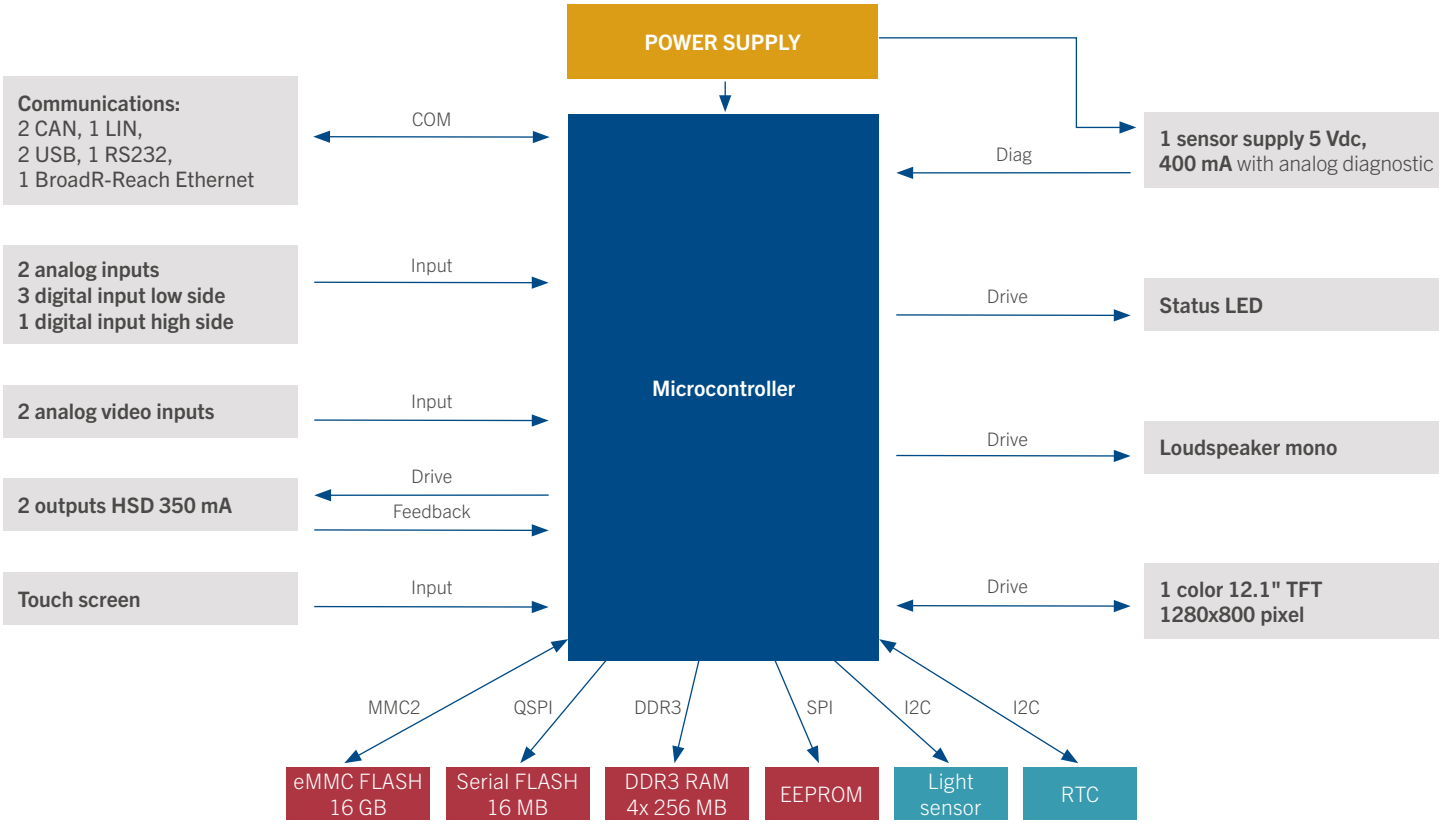
MTA SPEEDY CREATOR

MTA is offering a customized version of Android Studio (the worldwide integrated development environment for Google's Android OS) to enable customers to easily program dashboards and displays; MTA Speedy Creator extends the Android framework to provide access to common automotive interfaces and includes a widgets library for simplifying human machine interface definition.

MTA GATE

MTA Gate is a specific tool designed for EOL department and/or for technical service points use.

BLOCK DIAGRAM



SPECIFICATIONS

CHARACTERISTIC	DESCRIPTION
Operating voltage	
Nominal	12 and 24 Vdc
Supply voltage, permissible range	8÷32 Vdc
Power requirement	
Average power draw	24 W at 12 and 24 Vdc
Peak power draw (all loads)	64 W at 24 Vdc and 54 W at 12 Vdc
No load (inputs/outputs floating, Vref floating)	< 2.0 A at key on (12 Vdc) or <1.0 A at key on (24 Vdc)
With load (inputs/outputs active, Vref active)	<4.5 A at key on (12 Vdc) or <2.7 A at key on (24 Vdc) +2 x 350 mA (high side current) for both 12 and 24 Vdc +400 mA on sensor output +2 x 1.5 A on USB
Standby current	<2 mA at key off (12 Vdc and 24 Vdc)

Power output	
Power source for external sensors	5 Vdc ±5% max 400 mA / 2 W (max)
Thermal and IP characteristics	
Operating ambient temperature	-30 to +70 °C (-22 to 158 °F)
Storage temperature	-40 to +85 °C (-40 to 185 °F)
Ingress protection rate	IP66
Weight	
Total mass	2250 g (4.96 lbs)
Mounting location	
Suggested	Cab interior
Status LED	
High efficiency dual color LED	Red or green, software configurable
TFT	

CHARACTERISTIC	DESCRIPTION
Size	12.1" normally white
Form factor	16:10
Brightness	1000 cd/m^2 typical; 850 cd/m^2 minimum (at 20 °C)
Contrast ratio	600:1 minimum (at 20 °C)
Color depth	24 bits (3x8 bits per color)
Resolution	1280x800 pixels
Viewing area	261x163 mm
Viewing angles	-80~80°(H), -60~80°(V)
Backlight	With automatic and user programmable dimming function
Protective glass features	
Optically bonded (OCA)	✓
Anti-reflection finish	✓
Anti-fingerprint finish	✓
Touch screen	
Touch screen	Capacitive with multi-touch capability (up to 10 points). Detection compatible with light glove
Microcontroller	
Family and manufacturer	Texas instruments Jacinto 6 (DRA744x)
Clock frequency	1 GHz
Cores	2x A15, 2x M4
GPU	SGX544 core 3D Graphics accelerator + 2D G320 BitBLT
DSP	C66x digital signal processing core
Internal ROM	256 kB
Internal RAM with ECC	512 kB
External memory capability	
DDR3 RAM	1 GB
Flash EMMC	16 GB (application and operating system)
Serial flash (QSPI bus)	16 MB (boot process)
EEPROM (SPI bus)	32 kB (application configuration)
Real time clock	
Backed up by capacitors	Typical retention time >10 minutes, precision at extremes (-40, +85 °C) ±2.17 sec/day
Audio	
Internal speaker	Sound intensity of 85 ±3 dbA at 1 kHz measured at 10 cm from speaker on the back panel
Stereo output	Unbalanced, line level output, -3 dB BW: 15 Hz to 30 kHz RLOAD > 1 kΩ, 1.45 VPP; external amplifier required
Analog video inputs	
2x automatic PAL/NTSC	On Mini-BNC connectors
Input characteristics	As per analog video standard: 75 Ω, 1 Vpp
Other	
Ambient light sensor	Ambient light sensor for user-evaluated night and day view managing. No algorithm implemented
Battery voltage measurement	Accuracy: ±(8% +0.2 V) (-30÷70 °C)
Interfaces	
2x CAN bus	2.0B high speed (up to 500 Kbps) ISO 11898 – SAE J1939 capable Wake-up function CAN 1 termination: software selectable only when powered CAN 2 termination: not populated
1x LIN	LIN 2.2 Master Wake-up function
1x RS232	Up to 115 kbps only for DEBUG
1x BroadR-Reach	100 Mbps on 1 UTP
2x USB	USB 2.0 'type A' connector on the side, 1.5 A charging current
USB 2.0 on the rear Tyco connector, 1.5 A charging current	
Multifunction inputs	
2x analog 0-16 Vdc / 0-32 Vdc (software configurable)	- (Pull down 6.8 kΩ), ADC with 12 bit resolution - Accuracy 0-32 Vdc: ±(8%+0.2 V) (-30÷70 °C) - Accuracy 0-16 Vdc: ±(8%+0.1V) (-30÷70 °C) - No built in fault detection available
3x digital low side input	- Internal pull-up 6.8 kΩ, no wake function - No built in fault detection available
1x digital high side input	- Internal pull-down 6.8 kΩ, wake function on rising edge - No built in fault detection available
Power outputs	
2x high side outputs	- Max 350 mA with current sense. PWM up to 50 Hz, duty 2÷98%
Additional features	
Reverse polarity protection on power supply	✓
Reverse polarity protection on inputs	✓
Reverse polarity protection on outputs	✓
Rear button	Wake-up function possible, user programmable functionality
Materials and other mechanical characteristics	
Front frame	Plastic PC+ABS/TPS Black RAL 9005 painted
External cover glass	Mineral glass
Back side	Metal aluminum Black RAL 9005 painted
Chassis to environment air exchange	Gore valve (P/N GORE AMF200200)

CHARACTERISTICS

COMMUNICATION INTERFACES

2x CAN

The device provides 2 CAN ports capable up to 500 Kbps.
ISO 11898-1 and ISO11898-2 (CAN standard) – SAE J1939

CAN 1:
- PIN 1 CAN1-L
- PIN 2 CAN1-R
- 120 Ω termination software selectable. Wake-up functionality available
- Default speed set to 250 kBaud

CAN 2:
- PIN 3 CAN2-L
- PIN 4 CAN2-R
- No termination present on board. Wake-up functionality available
- Default speed set to 250 kBaud

1x BroadR-Reach Ethernet

The device provides 100 Mbit/s transmit and receive capability over a single Unshielded Twisted Pair (UTP) cable through this automotive Ethernet standard IEEE 802.3bw.

- PIN 5 ETH1-BR+
- PIN 6 ETH1-BR-
- 120 Ω UTP cable
- 100 BASE-T1, compliant with IEEE 802.3bw
- Wake-up capability unavailable

2x USB 2.0

A full speed USB on a type A connector is available on the side of the device, featuring a USB battery charger detection (BC 1.2 specification). A second USB 2.0 interface is available on the Tyco connector, configured as a standard downstream port. This interface has a 1.5 A charging current. USB short circuit to battery is currently unsupported for both interfaces. Charging currents are user programmable in EEPROM.

USB 1:
- Type 'A' connector on the side of the device
- USB 2.0 full speed
- 1.5 A charging current (default)

USB 2:
- On Tyco Superseal connector
- USB 2.0 full speed
- 1.5 A charging current (default)
- PIN 8: USB2_D+
- PIN 9: USB2_D-
- PIN 16: USB2_GND
- PIN 17: USB2_VCC

1x RS232

Port used only for debugging purposes. With speeds up to 115200 baud.

- PIN13 RS232_TX
- PIN12 RS232_RX
- PIN11 RS232_GND
- Speeds up to 115200 baud
- Managed by the operating system, unavailable for Android

1x LIN 2.2 Master

A local interconnect network port is available. A serial communication bus developed for cost sensitive automotive applications. This port acts as a Master. PIN 10: LIN communication pin.

- Use power ground for return signal. 1 kΩ pullup (to V_{BATT})
- LIN2.2A/SAE J2602. Up to 20 kb/s, set to 19.2 kb/s

DIGITAL AND ANALOG INPUTS

1x digital high side input

- PIN 14 corresponding to IN 1 on the Tyco connector, wake-up function can be configured on this pin.
- Logic level 0 when voltage input under 1 Vdc ±0.3 Vdc over the entire temperature range[3].
- Logic level 1 when voltage input over 5 Vdc ±0.6 Vdc over the entire temperature range[3].
- Pull-down of 6.8 kΩ to GND.
- Range from 0 to 32 Vdc.

3x digital low side inputs

- PIN 21 corresponding to IN 2 on the Tyco connector, no wake-up function associated to this pin.
- PIN 30 corresponding to IN 4 on the Tyco connector, no wake-up function associated to this pin.
- PIN 15 corresponding to IN 6 on the Tyco connector, no wake-up function associated to this pin.
- Logic level 1 when voltage input under 1 Vdc ±0.3 Vdc on both inputs over the entire temperature range[3].
- Logic level 0 when voltage input over 5 Vdc ±0.6 Vdc on both inputs over the entire temperature range[3].
- Pull-up of 6.8 kΩ to V_{BATT}.
- Range from 0 to 32 Vdc.

2x analog inputs

- PIN 22 corresponding to IN 3 on the Tyco connector.
- PIN 31 corresponding to IN 5 on the Tyco connector.
- 0-16 or 0 to 32 Vdc dynamic range, software configurable.

1x key input

- PIN 27 on the Tyco connector.
- Pulled down with a 15 kΩ resistor.
- Minimum voltage: 4.5 Vdc ±0.5 Vdc over the entire temperature range[3].

DIGITAL OUTPUTS

2 x digital high side outputs

- PIN 28 corresponds to OUT 1.
- PIN 29 corresponds to OUT 2.
- Max 350 mA with current sense. PWM up to 50 Hz, duty 2÷98%.
- PWM frequency and resolution in accordance with rise and fall time of the driver.
- Protected against short circuit and overload.
- Current sense with ±10% accuracy with load >50 mA.
- I_{MAX} 350 mA.
- R_{MIN} 35 Ω at 12 Vdc or 70 Ω at 24 Vdc.
- L_{MAX}: 50 mH; in case of PWM output management it's recommended an external protection of the output.
- An evaluation must be conducted prior to driving inductive loads outside of the stated specifications.

AUXILIARY SUPPLIES

1x AUX POWER output

An auxiliary power supply to power external sensors is provided at 5 Vdc (accuracy: ±5%), with 400 mA of maximum current, over the entire range of 8÷32 Vdc. The microcontroller monitors this power supply. The output is protected against short circuit to ground and to VBatt.

- PIN 20 corresponds to Vss.
- PIN 19 corresponds to GND_Vss.
- 5 Vdc ±5%, 400 mA over the entire range from 8 to 32 Vdc.

VIDEO

2x analog video inputs

These inputs accept analog video inputs. An automatic sensing mechanism is in place to detect NTSC or PAL video signals.

- MINI BNC 1: Video in 1.
- MINI BNC 2: Video in 2.
- Automatic NTSC/PAL switching.
- Single ended 75 Ω connections 1 Vpp, on board fixed termination.

3. Tamb -30 °C … +70 °C

AUDIO

1x stereo line output

'Audio Out' line level audio out. Conveys audio from internal audio sources to an external amplifier.

- PIN 23: AUDIO_OUT_R.
- PIN 32: AUDIO_OUT_L.
- PIN 24: AUDIO_GND.
- $V_{out} \leq 1.45\ V_{pp}$.
- $R_{LOAD} > 1\ k\Omega$ (external load).
- Frequency response -3 dB, from 15 Hz to 30 kHz.
- Single ended, AC coupled.

LOAD DUMP PROTECTION

Complete load dump protection is available for 12 Vdc and 24 Vdc systems:

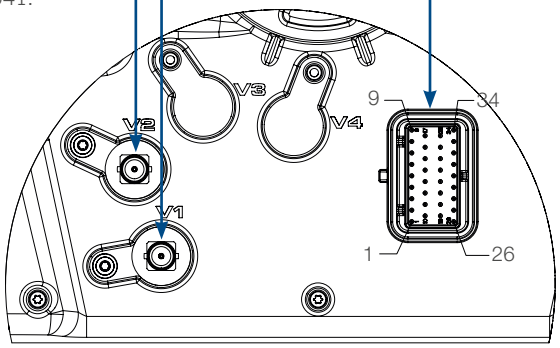
- Load dump test for 12 Vdc system (without centralized suppression):
Us = 101 Vdc; Ri = 1 Ω; Td = 400 ms; Tr = 10 mS.
Resistance to two pulses with a 60 second interval.
- Load dump test for 24 Vdc system (without centralized suppression):
Us = 200 Vdc; Ri = 2 Ω; Td = 350 ms; Tr = 10 mS.
Resistance to two pulses with a 60 second interval.

An evaluation must be conducted in case different test requirements are necessary.

CONNECTORS

Main connector
Tyco 34 pins, 3 keys, PN# Tyco 4-1437290-0.

Video inputs 1 and 2
Molex MINI BNC connector
PN# 73180-5041.



FUSES

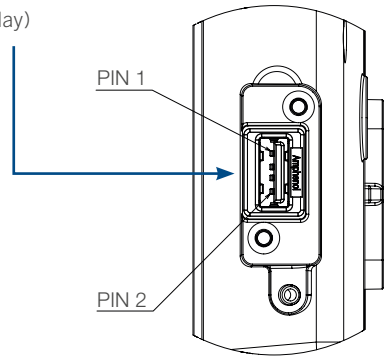
The fuse value suggested on the battery power supply is 7.5 A MiniVal.

VIDEO INPUT CABLES

The video input cable must be properly designed and produced to guarantee the video signal integrity and quality. MTA suggests to use a Leoni Dacar® 360 type cable (nominal cross-section 0.089 mm^2; impedance 75 Ω) or similar for automotive applications.

Depending on to the video camera, a specific connector must be selected (e.g. BNC 75 Ω, diameter 14.5 mm), which must be properly soldered and insulated with heat shrink material. Please contact MTA for proper connection suggestions.

USB (on the right side of the display)
Amphenol USB type A connector
PN# MUSBR-A111-R0.

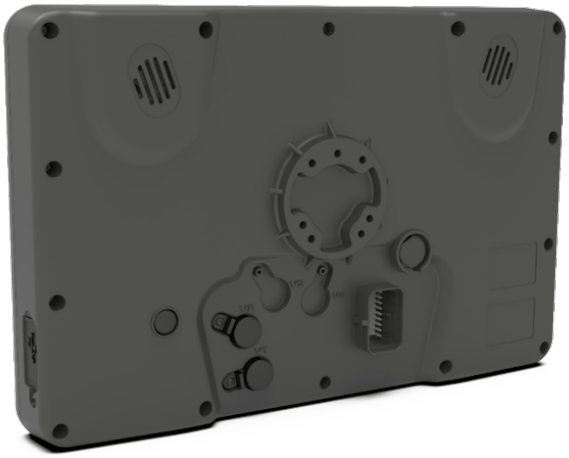


PINOUTS

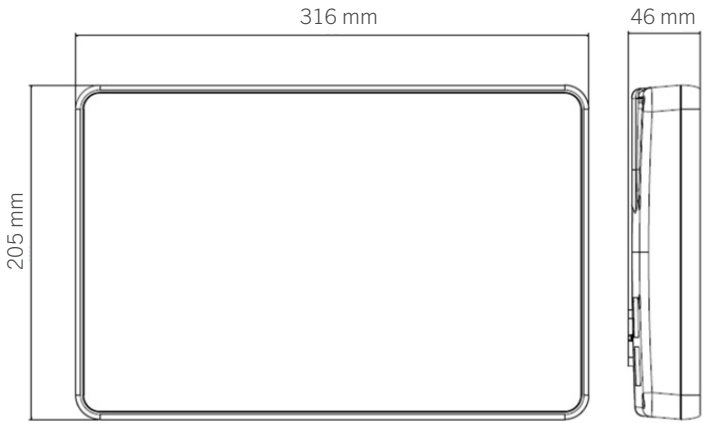
PIN	NAME	TYPE
1	CAN1_L	CAN1 bus low
2	CAN1_H	CAN1 bus high
3	CAN2_L	CAN2 bus low
4	CAN2_H	CAN2 bus high
5	ETH1-BR+	BroadR-Reach Ethernet1 Plus
6	ETH1-BR-	BroadR-Reach Ethernet1 Minus
7	- reserved -	- reserved -
8	USB2_D+	USB2 Plus
9	USB2_D-	USB2 Minus
10	LIN	LIN bus
11	RS232_GND	RS232 ground
12	RS232_RX	RS232 in
13	RS232_TX	RS232 out
14	IN_1	Digital input high side with wake-up function on rising edge
15	IN_6	Digital input low side
16	USB2_GND	USB2 ground
17	USB2_VCC	USB2 power
18	GND (terminal 31)	Battery ground
19	GND Vss	Sensor supply ground
20	Vss	+5 Vdc AUX power supply
21	IN_2	Digital input low side
22	IN_3	Analog input 0-16 Vdc or 0-32 Vdc
23	AUDIO_OUT_R	Audio output right
24	AUDIO_OUT_GND	Audio output ground
25	- reserved -	- reserved -
26	VB (terminal 30)	Battery power supply
27	VCTRL (terminal 15)	Ignition input
28	OUT_1	High side output 350 mA
29	OUT_2	High side output 350 mA
30	IN_4	Digital input low side
31	IN_5	Analog input 0-16 Vdc or 0-32 Vdc
32	AUDIO_OUT_L	Audio output left
33	- reserved -	- reserved -
34	- reserved -	- reserved -

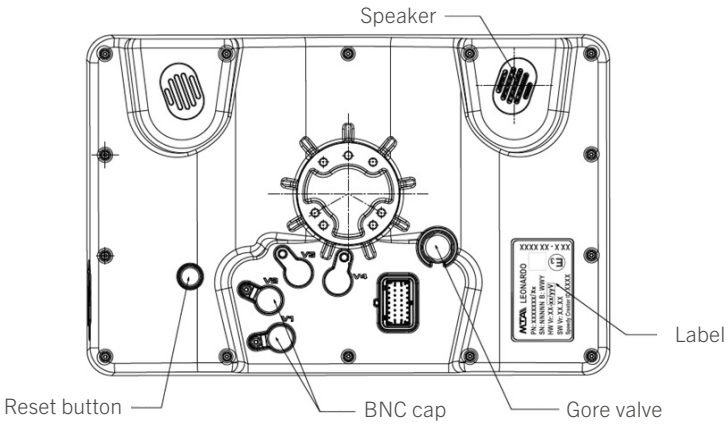
LAYOUT





DIMENSIONS





LCD

Features:

- diagonal 12.1";
- form factor 16:10;
- active area 261 x 163 mm;
- resolution 1280 x 800 pixels, 262k colors;
- portrait and landscape modes supported (see viewing angles below);
- contrast ratio: min 600:1;
- normally white;
- brightness: 1000 cd/m^2 (at 20 °C) typical, 850 cd/m^2 minimum;
- viewing angles: -80~80°(H), -60~80°(V);
- transmissive color TFT.

LCD APPEARANCE SPECIFICAION

The following specification defines the amount of unexpected defects on the LCD panel. Bright dots are pixels that are always on (never turn dark), dark dots are defined as pixels that are always black (never turn bright). The total amount of dark + bright is specified on the last line. If upon reception the TFT panel shows the following defects, it is rejected.

		Max number
Dot-defect	Bright dot-defect	N≤3 dot
	Dark dot-defect	N≤3 dot
	Total (bright+dark) dot	N≤5 dot

FIXING

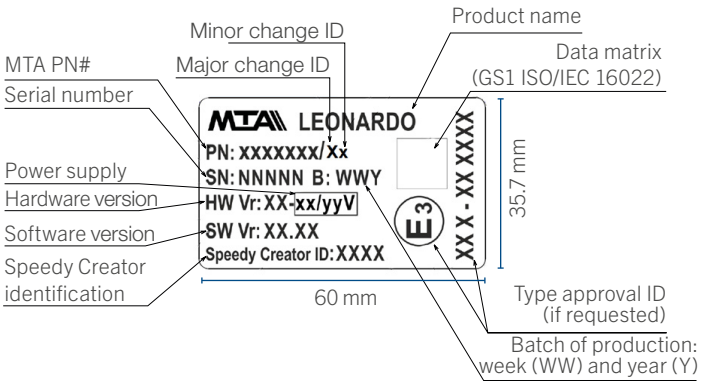
Fixing to the vehicle is obtained using a RAM mount system. Part number suggested: EC1414.
The datasheet for the mount is available, contact your MTA sales representative.

MATING CONNECTORS

The mating connector used for the main wire harness is a Tyco 34 pin Superseal.
The following parts are required for assembling a wiring harness connector:

- 34 pins, 3 key, connector PN#: C-2-1437285-3-05;
- terminal P/N: TE 3-1447221-3 (Ø 0.75 to 1.25 mm^2 conductor, Ø 1.8 to Ø 2.4 mm insulation);
- terminal P/N: TE 3-1447221-4 (Ø 0.5 mm^2 conductor, Ø 1.6 to Ø 2.2 mm insulation).

LABEL DETAILS



PRODUCT QUALIFICATION

MTA's reference test specification is TS50_16 (Rev. 6) which includes:

- environmental testing
- mechanical testing
- chemical testing
- electrical testing
- EMC testing
- CE type approval

Please contact MTA for details on any or all specific tests.

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